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# ON SURFACES AND THEIR CONTOURS

Roberto Pignoni

We prove two theorems concerning the global behaviour of a smooth compact surface $S$, without boundary, embedded in a real projective space or mapped to a plane. Our starting point is an analysis of the orientability properties of the normal bundle of a singular projective curve. Then we see how an excellent projection from $S$ to the Euclidean plane gives rise to integral relations linking the singularities of the apparent contour. Finally, given an embedding of $S$ in $RP^n$, we look at the discriminant $\Delta^*$ of a net of hyperplanes that intersects $S$ in a generic way, obtaining a characterization of $\Delta^*$ in terms of mod.2 cohomology invariants.

## 0. Introduction

In this paper we undertake a study of surfaces, developed from a global point of view by considering apparent contours and their duals. Our aim is to illustrate the interrelation between some features of these curves and the topology of the surface to which they belong. We are led, in this way, into a region that has been depicted as "a mixture of integral and mod.2 phenomena" [K].

The three sections of this article are meant to unfold different aspects of the subject.

Section 1 concerns general properties of curves with cusps and double points in a real projective plane. Some very appropriate remarks of the referee (unknown to us, but whose qualified criticism has been of great help in smoothing the style and improving the structure of this paper) have stimulated us to redesign this part. We have introduced a new and simple presentation of a few basic topological facts concerning singular projective curves. It may be of interest that the present approach leads to a straightforward proof of a slight refinement (Proposition 3) of a celebrated theorem of Möbius [Mö]. Our argument is essentially topological (a discussion of related aspects can be found in [Po]).

In Section 2 we apply to apparent contours our results on singular projective curves, making use of some Morse theoretical instruments introduced by Banchoff and Kuiper

[B-K]. Then we sacrifice a line in the target of our mappings and switch from projective to Euclidean plane. New possibilities are gained, due to the orientation properties that become available. We focus our attention on generic profiles of surfaces: it is well known that an *irreducible* contour in $R^2$ can be generated by a surface (which may, or may not, be orientable) of arbitrarily high genus (Fig.1). For this reason, at first sight there seems to be little hope of finding a significant connection between the number of irreducible components displayed by the contour and the genus of the surface.

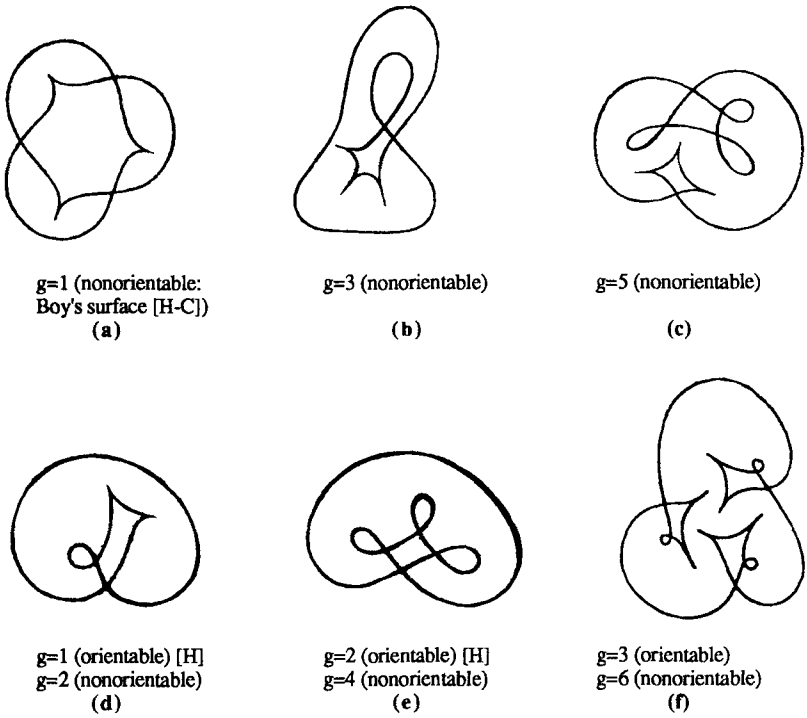



Fig.1

Our purpose is to show that such a connection exists, provided we count the components in the right way. Theorem 1 establishes a relation between the genus $g$ of the surface and the number of cusps, double points and irreducible components (suitably counted) presented by its generic apparent contour. It outlines also a new procedure, alternative to the one described in Proposition 7, enabling us to recover the Euler-Poincaré characteristic of a surface by looking at its oriented apparent contour in the Euclidean plane (our indebtment with the classical theory of the rotation number of a plane curve, due to Whitney [W1], is self-evident). For a first application of this result, see [B-D]; more consequences shall be discussed in a forthcoming paper [Pi2].

In Section 3 we consider surfaces embedded in $n$-dimensional real projective space. We intersect them with a generic linear net of hyperplanes, giving a cohomological description of the discriminant defined in this operation (Theorem 2). The information obtained in this way can be used in order to characterize, by means of duality arguments, the second Stiefel-Whitney number of the surface in terms of the cusps of a generic mapping with values in $RP^2$ (Corollary 2). The reader may suppose there is some kind of affinity between our approach to the topology of a surface embedded in $RP^n$, investigated by means of intersection with a net of hyperplanes, and some well known constructions of complex algebraic geometry. The theory of Lefschetz pencils, for instance, as developed by Andreotti and Frankel [A-F]. However, the technical foundations for the study undertaken in Section 3 rest rather on Looijenga's theory of generic families of functions [Lo] and on John Mather's results on generic projections [Ma].

The first formulation of some concepts introduced in this paper had been outlined in joint work with Alessandro Orlandi [O-P]. Partial results were presented, without proof, in [Pi1]. On that occasion we sketched a first survey of some literature connected with these subjects.

## 1. Singular Curves in a Projective Plane

Throughout this paper we will be concerned with a class of curves which may present singularities. They are described in the following

**Definition 1** *A compact connected subset* $\gamma$ *in* $RP^2$ *is a* curve with cusps and double points *if* $\forall x \in \gamma\ \exists U$ *open in* $RP^2$, *with* $x \in U$, *and a diffeomorphism* $\psi: U \to R^2$, *such that* $\psi(U \cap \gamma)$ *is* :

1) *either the set* $\{(x, y) \in R^2 \mid y = 0\}$;
2) *or the set* $\{(x, y) \in R^2 \mid xy = 0\}$;
3) *or finally the set* $\{(x, y) \in R^2 \mid x^2 - y^3 = 0\}$.

*We say that such a curve is* irreducible *if it can be parametrized by one smooth* ($C^\infty$) *map* $\varphi: S^1 \to RP^2$, *which is singular only when the image is a cusp point.*

In a parametrization of $\gamma$ we will assume that $S^1$ is *oriented* and it is assigned a base point $a$, such that $\varphi(a)$ is not an inflection point. If $p_1$ precedes $p_2$ in $S^1$ (that is, moving from $a$ in the sense prescribed by the orientation, we meet $p_1$ first) we will write $p_1 < p_2$.

If we remove from $\gamma$ every cusp and double point, we are left with a finite number of connected arcs, which are submanifolds of $RP^2$. Let's fix an orientation for the normal bundle of each of these arcs and see what happens near the singularities (Fig.2).

If $x$ is a *cusp* point, orientations have been assigned to the two different arcs which contain $x$ in their closure. We say they are *coherent* if, taken a small neighbourhood $U$ of

$x$ such that $\{U \setminus \gamma\}$ has two connected components, the positive direction always points towards the same connected component.

If $y \in \gamma$ is a *double point*, arising from a self-intersection of an irreducible component $\gamma_i$, take a smooth parametrization $\varphi: S^1 \to RP^2$ of $\gamma_i$. There are two points $p_1$, $p_2 \in S^1$, $p_1 \neq p_2$, such that $y = \varphi(p_1) = \varphi(p_2)$. We can choose two disjoint connected open subsets $I_1, I_2$ in $S^1$, with $p_i \in I_i$, such that $\varphi(I_1)$, $\varphi(I_2)$ are submanifolds of $RP^2$ only intersecting at $y$. We say an orientation of the normal bundles of the four arcs having $y$ in their closure is *coherent* (at $y$) if it can be induced, in a neighbourhood of $y$, by *global* orientations of the normal bundles of $\varphi(I_1)$ and $\varphi(I_2)$. An analogous definition can be given for a double point where two different irreducible components of $\gamma$ intersect.

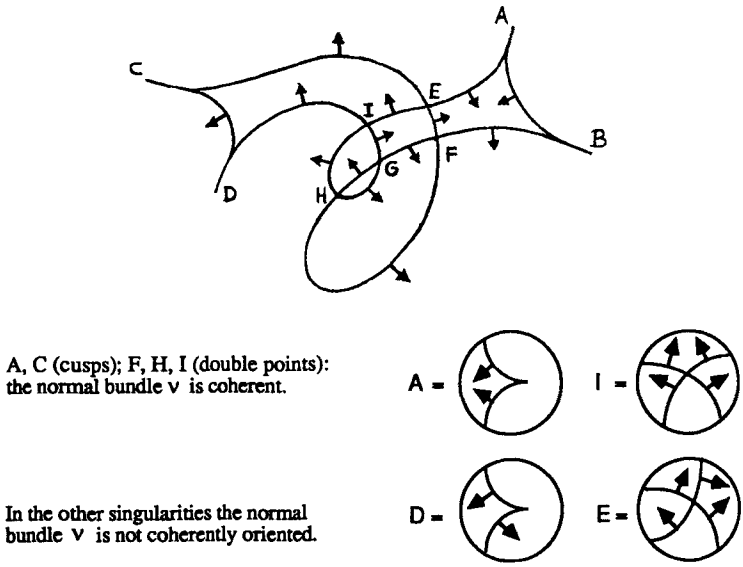



Fig. 2

**Definition 2** *A* curve with oriented normal bundle $\{\gamma, \nu\}$ *will mean a curve* $\gamma$ *together with an orientation of the normal bundle of each component of the complement of double points and cusps in* $\gamma$, *such that the orientation is coherent, in the above sense, across each cusp and double point.*

**Proposition 1** *Let* $\gamma$ *be a curve with cusps and double points in* $RP^2$, *and let* $\gamma_i$ *be its irreducible components. The normal bundle* $\nu$ *can be globally oriented if and only if each* $\gamma_i$ *is homotopic to zero.*

***Proof*** Let $r$ be an oriented projective line intersecting an irreducible component $\gamma_i$ in a generic way and let $\{p_1,..., p_m\}$ be the elements of $\{r \cap \gamma_i\}$. If we take a tubular neighbourhood $N$ of $r$ the orientation of $r$ induces an orientation of $\partial N$, as shown by Fig.3-left.

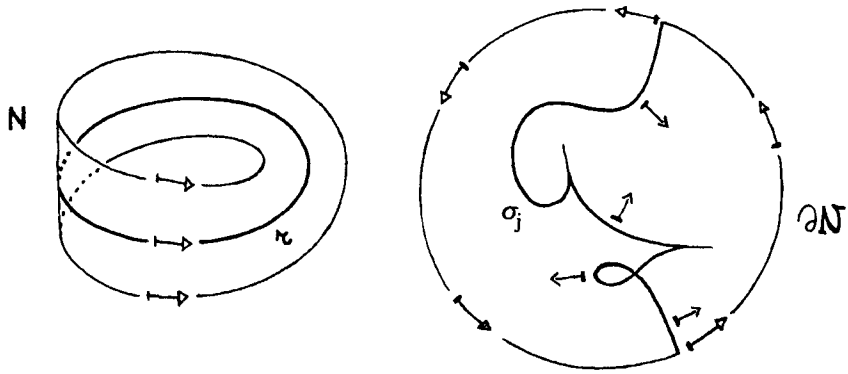



Fig.3

$N$ can be chosen in such a way that, in the affine plane $\{RP^2 \setminus r\}$, $\partial N$ is a simple oriented closed curve containing all the singularities of $\gamma_i$ in its interior and intersecting $\gamma_i$ transversally in $2m$ points. Let $\{\sigma_1,..., \sigma_m\}$ be the irreducible arcs of $\{\gamma_i \setminus N\}$. We can orient $v_{|\sigma_j}$ so that the orientation of its first fiber agrees with that of $\partial N$ (Fig.3-right). The resulting orientation at the last fiber of $v_{|\sigma_j}$ will then disagree with that of $\partial N$.

From this we see that $v_{|\gamma_i}$ has a global coherent orientation $\Leftrightarrow$ $m$ is even. A straightforward mod-2 homology argument shows that:

$m$ is even $\Leftrightarrow$ $\gamma_i$ is homologous to zero $\Leftrightarrow$ $\gamma_i$ is homotopic to zero. $\square$

Let's restrict ourselves to curves (with cusps and double points) *composed of a finite number of convex arcs*. We will call them "*convex-generic*". For this class of curves, we shall explicitate the link between the orientability of $v$ and the number of inflections of $\gamma$. $\mathbb{N}$ will denote the set of integers, endowed with the discrete topology.

**Definition 3** *Let* $\gamma$ *be irreducible and convex-generic. We say* i: $S^1 \to \mathbb{N}$ *is a* jump function *for* $\gamma$ *if there is a parametrization* $\varphi$: $S^1 \to RP^2$ *of* $\gamma$ (*associated to* i) *s.t.*:

1) $\mathrm{Im}(i) \subset \{1, 0, -1\}$ *and* $i(t) = 0 \Leftrightarrow \varphi(t)$ *is an inflection point*;

2) *for* t, t' *not corresponding to inflection points and such that* $t < t'$, *let's denote by* $f(t, t')$ *the number of inflections* $\varphi(x)$ *with* $t < x < t'$. *Then*: $i(t') = (-1)^{f(t, t')} i(t)$.

Suppose $\gamma$ admits a jump function. The correspondence: *parametrization of* $\gamma$ <-> *jump function on* $\gamma$ is many-to-one. It is clear that a jump function $i$ is constant on each

convex arc of $\gamma$, and its value "jumps" by two every time we pass an inflection point.

From now on, we shall denote by $\tau(p)$ the tangent line to $\gamma$ at $p$, defined also at cusp points, and by $f$ the number of inflections of the curve. We assume $\gamma$ to be irreducible.

**Lemma 1** *A jump function* i *for* $\gamma$ *exists* $\Leftrightarrow$ f *is even* .

***Proof*** The proof is immediate. Parametrize $\gamma$ and fix the value of $i$ on the base point. There is just one way to extend this value to the whole curve: when we get back to the base point, the function will be well defined if and only if $f$ is even. □

**Lemma 2** *The normal bundle* $\nu$ *is orientable* $\Leftrightarrow$ $\gamma$ *admits a jump function* .

***Proof*** **a)** => Fix an orientation $\omega$ on $\nu$ and a parametrization $\varphi$ of $\gamma$. We can define $i(t)$ as follows. Let $q = \varphi(t) \in \gamma$. We take a neighbourhood $U$ of $q$, which in a suitable affine chart is a disk centered at $q$, and such that $\{U \setminus q\}$ does not contain any double point, cusp or inflection point. Let's call $D_1,\ldots, D_k$ the connected domains of $\{RP^2 \setminus \gamma\}$.

If $q$ is not an inflection point, nor a singular point, we choose $U$ so that $\{U \setminus \gamma\}$ has two connected components and only one of them is a convex set, which lies in one of the $D_i$'s. We call $D_{conv.}(q)$ this $D_i$. We set $i(t) = 1$ if and only if the positive orientation of $\nu$ at $q$ points towards $D_{conv.}(q)$. At a cusp point $q = \varphi(t)$ we set $i(t) = 1$ if at any sufficiently close point $q'$ of $\gamma$ the oriented normal bundle $\nu$ heads towards $D_{conv.}(q')$; -1 otherwise. At a double point we delete the branch of the curve which is transversal to the tangent $\varphi'(t)$, and assign the value +1 or -1 in the same way.

**b)** <= $i: S^1 \to \mathbb{N}$ is given, and it is associated to a parametrization $\varphi: S^1 \to RP^2$ of $\gamma$. We orient the fibers of $\nu$ at points $q = \varphi(t)$ that are not inflections nor singular points, by requiring $\nu$ to point into $D_{conv.}(q)$ if and only if $i(t) = 1$. This orientation extends in a coherent way to all singularities and inflections of $\gamma$. □

**Definition 4** *We say that a jump function* i *is* associated *with a given orientation* $\omega$ *of* $\nu$ *if they are related in the way described in the proof of Lemma* 2, *points* **a)** *and* **b)**.

From what precedes, one immediately obtains (compare with [H-K, p.141]):

**Proposition 2** *Let* $\gamma$ *be a curve with cusps and double points, irreducible and convex-generic. Then:* f *is even* $\Leftrightarrow$ $\gamma$ *is homotopic to zero.* □

Proposition 2 and the proof of Proposition 1 can be a convenient starting point in order to obtain a finer version of a theorem of Möbius [Mö, S, H-K]. The argument is elementary in character, and has a topological basis.

**Proposition 3** *Let $\gamma$ be a simple closed curve in $RP^2$ which is not contractible to a point. Then $\gamma$ must have at least three inflection points. Furthermore, if r is a line intersecting $\gamma$ in a generic way and disconnecting $\gamma$, these inflections belong to more than one connected component of* $\{\gamma \setminus r\}$.

***Proof*** Call $m$ the number of intersections between $r$ and $\gamma$. Then $m$ is necessarily odd (see the proof of Proposition 1). We begin with the case $m = 1$. Take a "small" tubular neighbourhood $N$ of $r$, whose border $\partial N$ is a simple closed curve embedded in the affine plane $\{RP^2 \setminus r\}$ and intersecting $\gamma$ transversally in two points $q$ and $q'$. We may assume $\partial N$ to contain all inflections of $\gamma$ in its interior (as in the proof of Proposition 1). If $p$ is the unique intersection between $r$ and $\gamma$, the respective position of $\gamma$ and $\tau(p)$ in a neighbourhood of $q$ and $q'$ is shown in Fig.4-left. Endow $\{RP^2 \setminus r\}$ with an Euclidean structure. If $h$ is the height function in the direction normal to $\tau(p)$, it must have a maximum and a minimum on $\gamma$ inside $\partial N$. The sign of the curvature function changes at least 3 times as we go from $q$ to $q'$, hence $\gamma$ must have at least three inflection points.

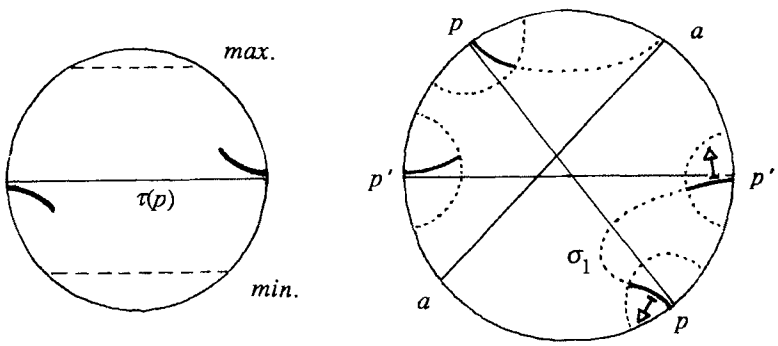



Fig.4

Let's consider what happens when $m > 1$ and $\{\gamma \setminus r\}$ has $m$ different connected components $\{\sigma_1, \ldots, \sigma_m\}$. We only need to prove the second statement of Proposition 3, because if two components contain inflections, these must be at least three. So, we are going to show that it is impossible that all the $\sigma_i$ but one be convex. We set $m = 2k + 1$ and carry on the proof by induction on $k$.

Let's see the case $k = 1$. Let's suppose there is just one component with inflections (call it $\sigma_1$). On $r$, $\sigma_1$ must have two distinct endpoints $p$ and $p'$ for otherwise $\gamma$ would be disconnected. After orienting the normal bundle $\nu$ of $\sigma_1$, we associate a jump mapping $i$: (0, 1) -> $\mathbb{N}$ to this (open) curve with oriented normal bundle (the definition being

patterned on that given for irreducible closed curves). Since $\sigma_1$ has an odd number of flexes and $\gamma$ is simple, in a neighbourhood of $p$ and $p'$ the portrait of the curve must be as in Fig.4-right (the straight lines representing tangents relative to $p$ and $p'$). But this makes it impossible to draw more than one convex arc joining the third intersection point $a$ with the other two, contradicting the hypothesis that $\gamma$ be a simple closed curve.

Now we are ready to prove our claim for $k > 1$. Again, let's suppose $\sigma_1$ is the only component of $\{\gamma \setminus r\}$ to display inflections, and let $p$ and $p'$ be its distinct endpoints. Call $\sigma_2$ the convex component starting at $p$. Between $\sigma_2$ and $r$ lies a convex domain $C$ (see Fig.5-left: again, the straight lines represent $\tau(p)$ and $\tau(p')$). In the closure of $C$ lies at least one connected domain of the same kind (the boundary being made of a convex arc $\sigma_i$ and a segment of $r$), but containing no convex components of $\{\gamma \setminus r\}$ in its interior. This arc $\sigma_i$ can be deformed by means of a regular homotopy into a new convex arc $\sigma'$ not meeting $r$ (Fig.5-right). In this way we reduce ourselves to the case of $m$ - 2 intersection points and the induction mechanism is released. □

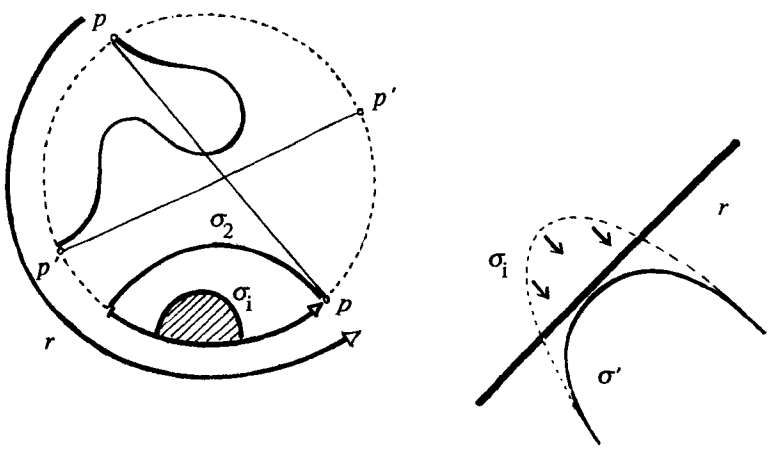



Fig.5

We will introduce now a different characterization for the orientability of the normal bundle $\nu$ of a singular curve $\gamma$. We don't assume $\gamma$ to be convex-generic, nor irreducible.

**Definition 5** *Let $\gamma$ be a singular projective curve. A function* f: $RP^2 \to \mathbb{N}$ *(as before, $\mathbb{N}$ denotes the set of integers, endowed with the discrete topology) will be called a* primitive function *for $\gamma$ if:*

1) f *is constant on the connected components* $\{D_1, ..., D_k\}$ *of* $\{RP^2 \setminus \gamma\}$ *and the value of* f(p) *jumps by two each time* p *crosses the curve transversely, at one of its regular points, passing from* $D_i$ *to* $D_j$;

2) *if we take a small convex neighbourhood* U *of a double point* q *of* $\gamma$, *such that* $\{U \setminus \gamma\}$ *has* 4 *connected components, then* f *assumes three distinct values on these components*;

3) *if* $p \in \gamma$, f(p) *is the average value of* f *for the* (2 *or* 4) *neighbouring regions* $D_j$.

**Proposition 4** $\gamma$ *has a primitive function* $\Leftrightarrow$ *its normal bundle* $\nu$ *is orientable.*

***Proof*** **a)** => Let $\varphi$ be a primitive function for $\gamma$. We can fix an orientation of the normal bundle $\nu$ as follows. Let us take a point $p \in \gamma$ that is not a double point, nor a cusp. Then $\exists\, j_1, j_2$ such that $p$ belongs to $\partial D_{j_1} \cap \partial D_{j_2}$ : if we take $q_1 \in D_{j_1}$ and $q_2 \in D_{j_2}$, $\varphi(q_1)$ differs from $\varphi(q_2)$ by two. Let's suppose $\varphi(q_1) = \varphi(q_2) + 2$: we assign to the fiber of $\nu$ in $p$ the orientation pointing towards $D_{j_1}$. In order to check that such a choice of orientation for each fiber of $\nu$ defines a global orientation $\omega$ of the bundle, not depending on the initial choice of $p$, it is sufficient to observe that points 1) and 2) of Definition 5 ensure that the orientations of the 4 arcs abutting on a double point fit together in a coherent way.

**b)** <= Choose an orientation $\omega$ for $\nu$. Fix a point $p$ not belonging to $\gamma$. If $q$ is another point of $\{RP^2 \setminus \gamma\}$, consider the two segments $A_1$ and $A_2$ joining $p$ and $q$. Orient each of them in the sense that goes from $p$ to $q$ and assign an index to each point $a$ at which $A_j$ and $\gamma$ intersect transversally: the index will be + 2 if, at $a$, the orientations of $A_j$ and of $\nu$ point to the same side of $\gamma$, - 2 otherwise. Now, the sum of all indexes relative to $A_1$ is equal to the sum of those relative to $A_2$. In order to see this, pick an irreducible component $\gamma_i$ of $\gamma$. Let $m_1$ be the sum of all indexes relative to $A_1 \cap \gamma_i$, $m_2$ the sum of those relative to $A_2 \cap \gamma_i$, and let's see that $m_1 = m_2$. Extend the orientation of $A_1$ to the whole line $r$ passing through $p$ and $q$ . Make a full loop along $r$, starting and ending at $p$ and assigning to each transversal intersection with $\gamma_i$ an index in the same way as before. The sum of all indexes will be $m_1 - m_2$. By applying the same argument exploited in the proof of Proposition 1, we see that this sum is zero, which entrains $m_1 = m_2$.

This enables us to define a primitive function $\varphi$. We start with $p$, setting $\varphi(p) = 0$ (notice that the choice of this value is totally arbitrary). If $q$ is a point of $\{RP^2 \setminus \gamma\}$, we take a segment $A$ joining $p$ and $q$, we orient it in the sense that goes from $p$ to $q$ and define $\varphi(q)$ to be the sum of all indexes relative to the transversal intersections between $A$ and $\gamma$, as before. We obtain a well defined function $\varphi$: $\{RP^2 \setminus \gamma\} \to \mathbb{N}$, which verifies points 1) and 2) of Definition 5 and can be extended to the points of $\gamma$ as prescribed by point 3). □

**Definition 6** *Let* $\varphi$ *be a primitive function for* $\gamma$, $\omega$ *an orientation of* $\nu$. *If they are related as in the proof of Proposition* 4 (*that is, when we cross* $\gamma$ *following the positive orientation of* $\nu$, *the value of* $\varphi$ *increases by two*) *we say they are* associated.

A primitive function determines a well defined orientation of $\nu$, but the converse is not true. There are infinitely many primitive functions associated with the same orientation $\omega$ of $\nu$ and any two of them differ by a constant. This is an easy consequence of the construction used in the proof of Proposition 4 (part **b**)). We have shown:

**Corollary 1** *If* $\nu$ *is orientable, there are two equivalence classes of primitive functions for* $\gamma$. *Two equivalent primitives differ by a constant, so that each primitive function is determined by its equivalence class and by its value at one point of* $RP^2$. □

Two primitives are equivalent ⇔ they are associated to the same orientation of $\nu$ ⇔ they differ by a constant. If $\varphi$ is a primitive for $\gamma$ and $a$ is an integer, $\varphi + a$ is a primitive function of the same class; $-\varphi$ is a primitive function belonging to the other equivalence class.

We have just remarked that there are infinitely many primitives associated with a curve with oriented normal bundle $\{\gamma, \nu\}$. However, if $\gamma$ *is convex-generic* it is possible to assign to $\{\gamma, \nu\}$ a special primitive function $\chi_\gamma$ which is defined in a canonical way.

Let's see how. At first, suppose $\gamma$ irreducible and take a jump function $i: S^1 \to \mathbb{N}$, associated with $\nu$'s orientation (Definition 4) and with a parametrization $\varphi: S^1 \to RP^2$ of $\gamma$. If $p \in RP^2$, let $\mathcal{L}(p)$ denote the pencil of lines of center $p$.

**Definition 7** *We set* $\mathbf{S}(p) = \{t_1, \ldots, t_\lambda \in S^1$ *such that* $\tau(\varphi(t_i)) \in \mathcal{L}(p)\}$. *Then*

$$\chi_\gamma(p) = \sum_{t \in \mathbf{S}(p)} i(t)$$

The meaning of this new function is illustrated by Fig.6. The value of $\chi_\gamma(p)$ is equal to the sum of as many terms +1 as there are points $q \in \gamma$ at which the tangent line $\tau(q)$ spans $p$, with the positive orientation of $\nu$ in $q$ pointing towards $D_{\text{conv.}}(q)$ (remember the proof of Lemma 2); and of as many terms -1 as there are points $q$ with tangent $\tau(q)$ spanning $p$, but at which it is the negative orientation of $\nu$ that heads towards $D_{\text{conv.}}(q)$. The value of an inflection point $q$, with $\tau(q)$ spanning $p$, is 0. A cusp point $q$, whose tangent spans $p$, counts +1 when in any sufficiently close point $q'$ of $\gamma$ the oriented normal bundle $\nu$ heads towards $D_{\text{conv.}}(q')$; -1 otherwise. Finally, in a double point we delete the branch of the curve which is transversal to the tangent, and assign the value +1 or -1 in the usual way.

If $\gamma$ has $h$ irreducible components $\{\gamma_1, \ldots, \gamma_h\}$ whose normal bundles have all been oriented, we can extend the above definition by setting: $\chi_\gamma = \sum_{i=1}^{h} \chi_{\gamma_i}$

**Proposition 5** $\chi_\gamma(p)$ *is a primitive function for* $\gamma$.

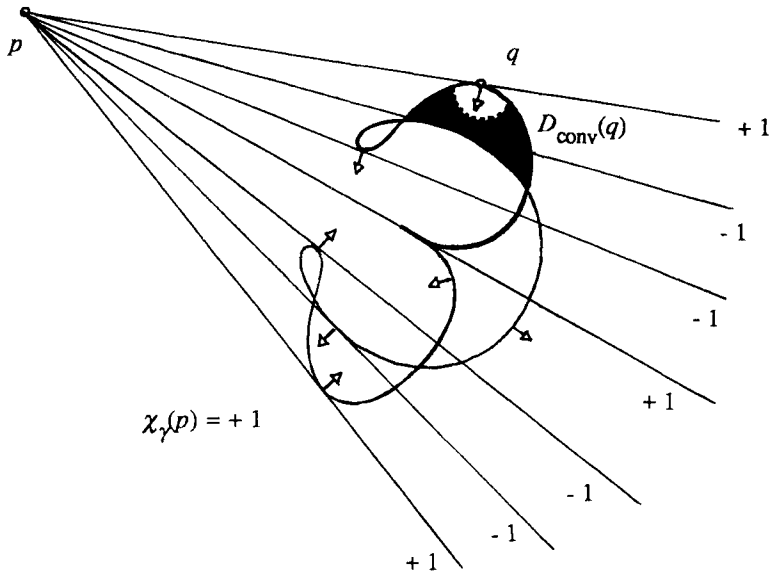



Fig.6

***Proof*** We are dealing with an integral valued function. Let's see that its value does not change as $p$ varies within the same connected component $D_j$ of $\{RP^2 \setminus \gamma\}$. We need to show that $\forall\ a, b \in D_j$ one has $\chi_\gamma(a) = \chi_\gamma(b)$. We will prove this statement assuming that the two points abut a segment $ab \subset D_j$ not lying on any tangent to $\gamma$ and such that either: **1)** $ab$ does not intersect any tangent to an inflection point, or: **2)** there is just one such intersection (at an interior point of $ab$). In fact, every couple of points in $D_j$ can be connected by a polygonal path whose segments are of this kind, the only exception occurring when one of the two points lies on the tangent to an inflection. And the arguments we will give still apply (with minor adjustments) to this case.

So let's see how things work in case 1). Pick a line $l \in \mathcal{L}(a)$, such that $l = \tau(p)$ for some $p \in \gamma$. Parametrize $\gamma$, take the two inflection points immediately preceding and following $p$ and call them $f_1$ and $f_2$, respectively. Let $C$ be the arc of $\gamma$ containing $p$ and going from $f_1$ to $f_2$. If we move along $C$ starting from $p$ and going in the proper direction, the tangent $\tau(q)$ will keep intersecting $ab$ until it gets to $b$ (which happens before $q$ reaches the inflection). $i$ remains constant and the contribution of the arc $C$ to $\chi_\gamma(x)$ is the same for all $x \in ab$. This argument can be repeated for all tangents $l$ drawn from $a$ to $\gamma$, and can be applied also when the component of $\gamma$ containing $p$ has no inflections. Case 1) is proved.

Now suppose there is a line $\tau(p)$ tangent to an inflection point $p$ and intersecting $ab$ at a point $c$ (case 2). Again, let's take an open arc $C$ of $\gamma$ containing $p$ and joining the two

closest inflections $f_1$ and $f_2$. Let's start from $p$ and move towards $f_1$. The tangent $\tau(q)$ sweeps $ab$ from $c$ to one of its endpoints (let's say $a$). If we go from $p$ towards $f_2$, we sweep $ca$ once again, but now the contribution to $\chi_\gamma(x)$, for $x \neq c$, has the opposite sign because the value of the jump function $i$ changes after an inflection. In conclusion, the contribution of the arc $C$ to $\chi_\gamma(y)$ is the same for all $y \in ab$ and it is always equal to zero. (The argument continues to hold if $f_1 = f_2$.)

Now we should check that $\chi_\gamma$ satisfies the second part of point 1), as well as points 2) and 3) of Def. 5: this follows from the definition of $\chi_\gamma$ and we leave it to the reader. □

The reader should have noticed that, as a primitive function, $\chi_\gamma$ is not associated to the orientation fixed on $\nu$ but to the opposite one. If we know the value of $\chi_\gamma(p)$ for one point $p \in \{RP^2 \setminus \gamma\}$ we don't need to draw any more tangents in order to determine its value for another point of the plane. When the point $x$ crosses $\gamma$, jumping from a domain $D_{j_1}$ to a bordering region $D_{j_2}$, the value of $\chi_\gamma$ varies by two: the function decreases if and only if the normal bundle of the curve at the crossing point heads towards $D_{j_2}$ (that is, if and only if $x$ crosses $\gamma$ following the positive orientation of $\nu$).

## 2. Oriented Apparent Contours

Take a compact surface $S$, connected and without boundary. If $f: S \to RP^2$ is a generic mapping, its critical point set is the union of $s$ circles $C_1, \ldots, C_s$ embedded in $S$. They constitute the *fold curve* of $f$ and their images are the irreducible components of the apparent contour $\gamma$, which is a curve with cusps and double points [W2, Ma]. We are going to see that $f$ determines, in a "canonical" way, two distinct primitives for this curve. They belong to different equivalence classes, hence their sum must be equal to a constant. We will realize that this constant has a topological meaning, being equal to *the Euler-Poincaré characteristic* $\chi(S)$ of the surface.

Let's introduce the first "canonical" primitive function associated with the contour.

**Definition 8** *We shall indicate with* $m: RP^2 \to \mathbb{N}$ *the function associating to every* $p \in RP^2$ *the number* $m(p)$ *of the points that lie in the fiber* $f^{-1}(p)$.

If $\{D_1, \ldots, D_k\}$ are the connected components of $\{RP^2 \setminus \gamma\}$, $m(p)$ is constant as $p$ varies in $D_j$, with $j$ fixed. We can check immediately that in fact we have defined a primitive function for the curve $\{\gamma, \nu\}$. By applying the results of Section 1, we obtain:

**Proposition 6** *Every irreducible component* $\gamma_i$ *of the apparent contour* $\gamma \subset H$ *has an orientable normal bundle and therefore is homotopic to zero.* □

**Definition 9** *We will refer to the orientation* $\omega$ *associated with* m (*Definition* 6) *as "the orientation of* $\nu$ *determined by* f", *and to the couple* $\{\gamma, \nu\}$ *with normal bundle oriented in this way as the* oriented apparent contour *of* S.

In the following, when referring to the contour $\gamma$ of a map we will make the implicit assumption that it is endowed with its canonical orientation $\omega$. If a component $\gamma_i$ of an apparent contour displays at least one cusp point, one sees from the local structure of the singularity that the only orientation fulfilling our criterion is the one pointing towards the "interior" of the cusp (as in Fig.2-A). Therefore, if each $\gamma_i$ presents cusps we can detect at once the right orientation for $\gamma$, without any need to know the value of $m(q)$. One more consequence of orientability is that we can never find, in a component $\gamma_i$ of the contour, an odd number of inflection points lying between two cusps (remember the proofs of Lemma 1 and Lemma 2, Section 1).

And now we come to the second "canonical" primitive function determined by a contour $\gamma$. The curve is not, in general, convex-generic. Nevertheless, for almost any choice of $p \in RP^2$ the pencil of projective lines which have $p$ in common describes, in a suitable affine chart $U$, all level curves of a linear function $L$ enjoying the property that $L \circ f$ is nondegenerate over $f^{-1}(U)$ [Ku, L2]. In a neighbourhood of a critical point of $L \circ f$ a small arc of the fold curve of $f$ is projected to a corresponding arc of $\gamma$, which is either a convex branch or contains a cusp point. This shows that for almost every $p \in \{RP^2 \setminus \gamma\}$ it is still possible to compute $\chi_\gamma(p)$ as in Definition 7. Our next step will be to establish that $\chi_\gamma$ can be extended to a primitive function $\psi$ defined on all of $RP^2$, thus obtaining two "canonical" primitives associated with $f$: $m$ and $\psi$. At the same time we are going to see that they belong to different equivalence classes, so that their sum $m + \psi$ is equal to a constant, and that this constant is the Euler-Poincaré characteristic $\chi(S)$ of the surface.

**Proposition 7** *If* $\gamma$ *is the apparent contour of a generic projection, the primitive function* $\psi = \chi(S) - m$ *coincides with* $\chi_\gamma$ *at all points for which the last is defined.*

***Proof*** We apply to maps to the projective plane an idea of [B-K]. Choose a point $p \in \{RP^2 \setminus \gamma\}$ for which $\chi_\gamma$ is defined, and consider the fiber $f^{-1}(p) = \{q_1, \ldots, q_{m(p)}\}$. As usual, we call $\mathcal{L}(p)$ the pencil of projective lines of center $p$. We define a smooth mapping $\phi$: $\{S \setminus f^{-1}(p)\} \to RP^1$ by associating to each point $x \in S$ the line of $\mathcal{L}(p)$ on which $f(x)$ lies. We study the function $\phi$ by means of the method developed by Banchoff and Kuiper in their treatment of the geometrical class and degree of a surface in three-space [B-K, Lemma 2.1, p.565]. Let's sketch a description of this method.

The starting point is the observation that the "Morse equality" $\sum_j (-1)^j \mu_j = \chi(M)$,

relating the Euler-Poincaré characteristic of a compact smooth manifold $M$ to the numbers $\mu_j$ of critical points of index $j$ of a Morse function on $M$, can be extended to the case of a smooth nondegenerate function from a compact manifold $M$ to the circle $S^1$.

The next step consists in removing from the surface $m(p)$ small open "round" neighbourhoods, each of them being the counter-image, through $f$, of a convex neighbourhood of the plane $RP^2$, centered at $p$. In this way, after getting rid of these neighbourhoods of the points of the fiber $f^{-1}(p) = \{q_1,..., q_{m(p)}\}$, one can construct a new surface $S'$ by identifying the "antipodal" points in the boundary of each round neighbourhood. The Euler-Poincaré characteristic of $S'$ is equal to $\chi(S) - m(p)$. Now, $\phi$ can be viewed as a nondegenerate function defined on the whole of $S'$, and the value of $\chi(S)$ equals the alternate sum of the numbers of critical points of $\phi$ of the same index.

The sign that must be attributed to each critical point can be determined as in Fig.6 (which, however, does *not* represent an oriented apparent contour!), by considering the mutual position of the line of the pencil and of $D_{conv.}(q)$ at the tangency point $q \in \gamma$. This shows that the alternate sum is equal to $\chi_\gamma(p)$. In this way, we get the following relation:

$$\chi(S) - m(p) = \chi_\gamma(p)$$

Since $m(p)$ remains constant as $p$ varies in the same connected component $D_j$ of $\{RP^2 \backslash \gamma\}$ and the "good" points are dense, the function $\chi_\gamma$ may be extended, by continuity, to all of $D_j$. □

***Remark* 1** The relation we have just proved: $\chi(S) = \chi_\gamma(p) + m(p)$, has analogues for mappings from an even-dimensional manifold to $R^2$ [L2] and also for generic mappings $f$: $M \to N$, where $M$ and $N$ are closed oriented surfaces. In the last case, the role of our function $\chi_\gamma$ is played by the "double tangent winding number" of the apparent contour relative to a suitable vector field [C2]. A global relation of a different kind can be obtained (for the same class of maps) by considering the "local degree" of $f$ at cusp points (see [Q], or [F-I] for the extension to oriented manifolds with boundary).

Corollary 1 and Proposition 7 imply that the knowledge of *both* $\omega$ and $m(p)$ (for at least one $p \in H$) is necessary and sufficient in order to identify $S$, starting from one of its generic apparent contours $\gamma$ in the projective plane. Of course, if $\chi(S)$ and $m(p)$ (for $p \notin \gamma$) are both even, this recognition will be possible only up to orientability (Fig.1-d,1-e,1-f).

That the knowledge of $\omega$ is necessary, it can be seen by means of a simple example of two non-homeomorphic connected surfaces (a sphere and a double torus) embedded in $R^3$ and projected orthogonally to a plane (Fig.7). They give rise to the same apparent contour $\gamma$. The necessity of knowing $m(p)$ follows from the simple observation that in an affine chart of $RP^3$ the hyperplane $\{z = 0\}$ and the two-sheeted hyperboloid defined by the equation $x^2 + y^2 - z^2 + 1 = 0$ may determine, when projected to a generic hyperplane, the same empty contour.

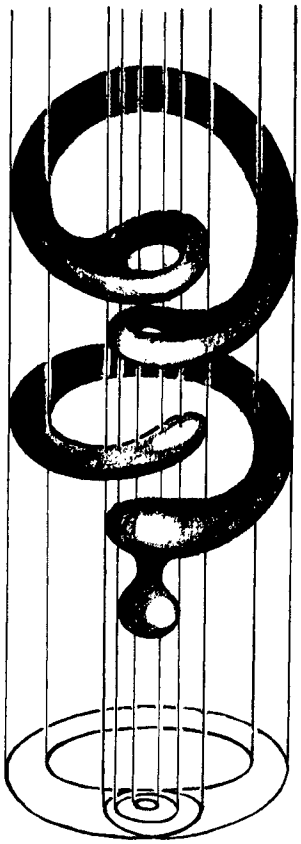


Fig.7

It must be remembered that there are limitations to the amount of information obtainable from an oriented apparent contour. As it has just been recalled we cannot distinguish, in general, between the contour of a torus and that of a Klein bottle. But even restricting ourselves to surfaces embedded in $R^3$, it is a hopeless effort to try to deduce from an oriented apparent contour nonambiguous indications about the way in which different "sheets" of the surface are glued together along the fold curve (Fig.8). This phenomenon was originally pointed out in a famous example, usually referred to as "Milnor's paisley" [P, p.11]. The first complete discussion of Milnor's curve came together with the classification of all non-equivalent immersions of the disk $D^2$ into the Euclidean plane extending a given immersion of $\partial D^2$ [Bl] (the question is related to the "Picard-Loewner problem" [Ti2]). Further developments concerned immersions of more

general surfaces into the plane [Tr, B] and "polymersions" to $R^2$ [M] or to the 2-sphere [F1] (a "polymersion" is an immersion which may have branch points). These results made it possible to classify, by means of combinatorial techniques, the different ways in which the neighbourhoods of the fold curves of a surface can be glued together so that they give rise to nonequivalent projections to $R^2$ or $S^2$ with the same oriented apparent contour [F-T1, F-T2, Wi]. Later work dealt with polymersions into a closed oriented surface of arbitrary genus [F2, E-M] and with the generalization of this approach to maps with non-orientable source or target [Ez]. Two recent contributions are [C-W] and [C1].

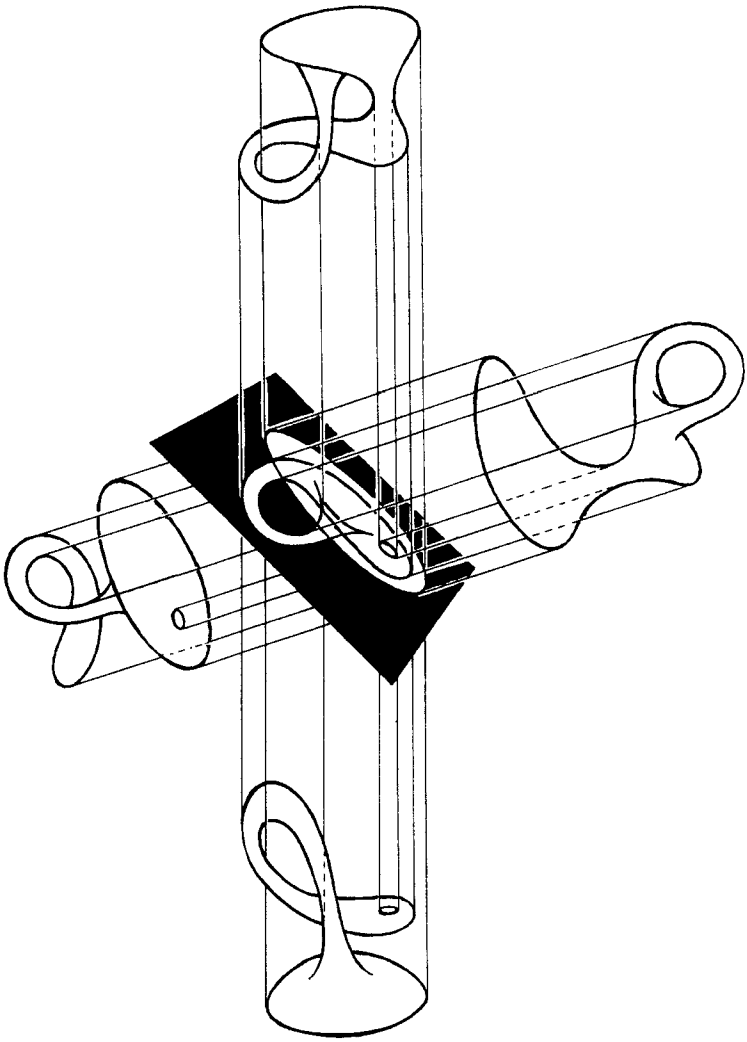


Fig.8

Now let's consider apparent contours in the *Euclidean plane*. Take a compact (connected) surface $S$ without boundary and a generic smooth mapping $f: S \to R^2$. We indicate with $\gamma_1, \ldots, \gamma_n$ the irreducible components of the oriented apparent contour $\{\gamma, v\}$. The canonical orientations of $v$ and of $R^2$ determine the sense according to which each $\gamma_i$ has to be parametrized. Let's fix a set of (nonsingular) base points $A = \{a_1, \ldots, a_n\}$, with $a_i \in \gamma_i$. If $p \in \{R^2 \backslash \gamma_i\}$, we denote by $w(p)$ the *winding number* of $\gamma_i$ with respect to $p$ (a quick way to compute it is explained in [Ti1, L3]).

**Definition 10** *Let* $c_1, c_2$ *be two points of* $\{R^2 \backslash \gamma_i\}$ *that are very close to* $a_i$ *and lie on different sides with respect to the curve. We define (compare with* [Ti1]) $\mu(a_i) = w(c_1) + w(c_2)$ *and assign a global index* $\mu(A)$ *to* $\gamma$ *by setting* $\mu(A) = \sum_{i=1}^{n} \mu(a_i)$.

Notice that $\{R^2 \backslash \gamma_i\}$ has one unbounded component, say $U$. If we choose $a_i$ on $U$'s border, the index $\mu(a_i)$ will be equal to -1 if the positive orientation of $v$ at $a_i$ points towards $U$, to +1 if it heads in the opposite direction. This means that if we make such a choice for the set $A$ of base points we attribute *a positive or negative sign to every irreducible component of the contour.*

The introduction of the set $A$ of base points enables us to assign a value $\eta(Q)$ to every double point $Q$ of $\gamma$. In the first place, if $Q$ arises from the intersection of two *different* irreducible components of the contour we set $\eta(Q) = 0$. If $Q$ is a point of self-intersection of the same component $\gamma_i$ we may define $\eta(Q)$ as in [W1]: we choose a parametrization $\varphi: S^1 \to \gamma_i$ with base point $a_i$, coherent with the orientation of the normal bundle and singular only when the image is a cusp point. We take $t_1, t_2$ with $t_1 < t_2$, such that $\varphi(t_1) = \varphi(t_2) = Q$, and set $\eta(Q) = +1$ if the couple $(\tau_1, \tau_2) = (\varphi'(t_1), \varphi'(t_2))$ induces the *opposite* orientation with respect to that of $R^2$. We define $\eta(Q) = -1$ if the two orientations agree; of course these definitions rely on our choice of $a_i$.

**Definition 11** *We shall denote by* N(A) *the sum of all values obtained in this way:* $N(A) = \sum \eta(Q)$ *(for all double points* Q *of the contour).*

**Theorem 1** *Let* c *be the number of cusps of* $\gamma$ *and* g *the genus of* S. *Let* $\varepsilon(S)$ *be equal to* 1 *if* S *is orientable and to* 2 *if it is not. We have:*

$$g = \varepsilon(S)\,[1 + \frac{c}{2} - \mu(A) - N(A)]$$

***Proof*** Let's pick an irreducible component $\gamma_i$ of $\gamma$. We define its *rotation number* $\rho(\gamma_i)$ by approximating the curve as follows. For each cusp $p = f(q)$, we consider the

component $C_i$ of the fold curve on which $q$ lies. We take, for a simply connected neighbourhood $V \subset S$ of $q$, a small tubular neighbourhood of $V \cap C_i$ in $S$ and call it $N_q$.

Let $\sigma_q$ be a never vanishing section of $N_q$, with image in the "interior of the cusp", that is, in the connected component of $\{N_q \setminus C_i\}$ which is mapped one-to-one to $R^2$ in a neighbourhood of $q$ [L3, p.166] (the dark component, in Fig.9). We approximate $\gamma_i$ with a new curve coinciding with $f \circ \sigma_q$ near $q$ and equal to $\gamma_i$ out of a small neighbourhood of $q$ [L3, p.161].

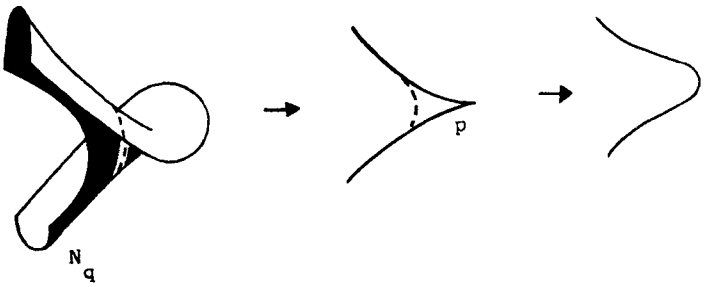



Fig.9

Once we have done so for all cusps in $\gamma_i$, we define $\rho(\gamma_i)$ as the "rotation number" [W1], divided by $2\pi$, of the curve $\gamma_i'$ that has been obtained by "smoothing off" all the cusps. The rotation number of $\gamma_i'$ can be seen as the degree of the map $\lambda$: $S^1 \to S^1$ obtained from a regular parametrization $\varphi$: $S^1 \to R^2$ of the curve $\gamma_i'$, *coherent with the orientation of* $v_{|\gamma_i}$, by setting $\lambda(x) = \{$the unit tangent vector to $\gamma_i'$ at $\varphi(x)\}$ (Fig.10).

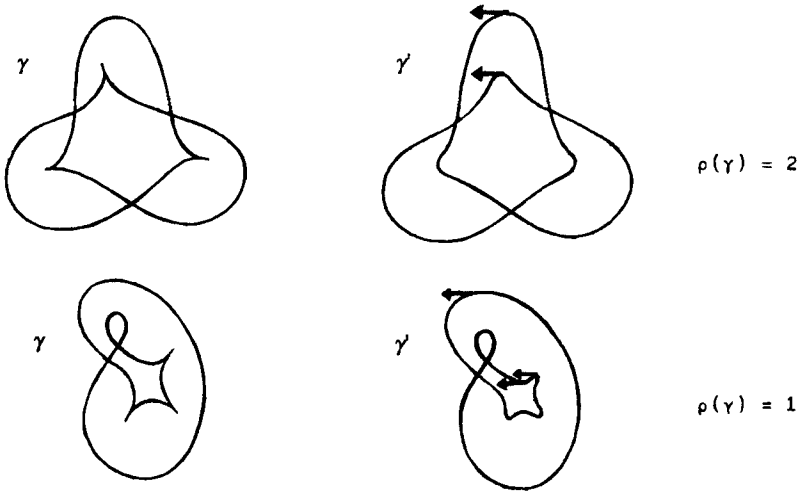



Fig.10

It is fairly obvious that, provided the approximation to the fold curve is $C^{\infty}$-small, our definition of $\rho(\gamma_i)$ does not depend on the particular approximating curve that is constructed.

We define $\rho(\gamma)$ as $\sum_{i=0}^{n}\rho(\gamma_i)$. This number is invariant under regular homotopies (or "deformations") of the curve $\gamma$ [W1, pp.278-279].

By means of a diffeotopy of the plane, we can construct a diffeomorphism $\psi$: $R^2$ -> $R^2$ that "bends" $\gamma$ in a neighbourhood of its cusp points and makes all the tangents to the cusps parallel to the same direction (Fig.11). In this process, the rotation number of the contour will not change. Now, embed $R^2$ into a projective plane and call $p$ the point at $\infty$ at which the tangents to the cusps intersect. If $\gamma^*$ is the contour of $\psi \circ f$, we have $\rho(\gamma^*) = \rho(\gamma)$. Considering one of the components of $\gamma^*$, say $\gamma_i^*$, and computing $\chi_{\gamma_i^*}(p)$ (Definition 7) we get $\chi_{\gamma_i^*}(p) = 2\rho(\gamma_i^*) - c_i$, where $c_i$ denotes the number of cusps of $\gamma_i$ (Fig.11-right).

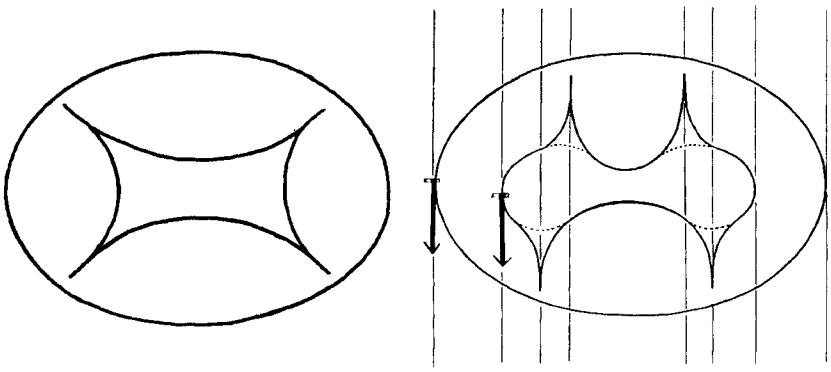


Fig.11

Summing up over all components of $\gamma^*$ and applying Proposition 7:

$$2\rho(\gamma^*) - c = \chi(S)$$

We recall that $\rho(\gamma)$ is equal, by our definition, to $\sum_{i=0}^{n}\rho(\gamma_i)$, and for each $i$ the following relation [W1, Ti1] is verified:

$$\rho(\gamma_i) = \mu(a_i) + \sum \eta(Q)$$

(the sum is computed *over the self-intersections of the component* $\gamma_i$). At last, summing up for all double points $Q \in \gamma$ and all irreducible components, we get:

$$2\mu(A) + 2N(A) - \chi(S) = c$$

The statement of Theorem 1 immediately follows. □

## 3. Generic Embeddings of Surfaces in $RP^n$

In Section 1 we described some properties of convex-generic projective curves with cusps and double points. Let's examine a way to produce this kind of objects. Consider a smooth compact connected surface $S$ without boundary. Realize it as a submanifold of $RP^n$ by means of an embedding $\varphi$: $S \to RP^n$, then choose an $(n - 3)$-hyperplane $C$ not intersecting $\varphi(S)$. The $(n - 1)$-dimensional hyperplanes that have $C$ in common determine a *linear net of hyperplanes* $\Sigma^*$, of *center* $C$. It's a two-dimensional projective space. The set of all elements of $\Sigma^*$ not transversal to $\varphi(S)$ is called the *discriminant* of the net.

A second projective plane $\Sigma$, dual to $\Sigma^*$, can be obtained by considering the family of all $(n - 2)$-hyperplanes containing $C$. The map $\pi$: $S \to \Sigma$ associating to each point $x \in S$ the element $\pi(x) \in \Sigma$ that contains $\varphi(x)$ is called the *projection* of center $C$. In this way we are given two operations, dual to each other: projection of center $C$, and *intersection with the net of hyperplanes* containing $C$. They can be carried out under specific genericity conditions that we are going to describe.

Let $N$ be an $(n - 1)$-hyperplane passing through a point $y \in \varphi(S)$. $N$ is said to have *contact of type* $A_k$ with $\varphi(S)$ at $y = \varphi(z)$ if, given an affine chart $(U, \psi)$ with $y \in U$, $\psi(y) = 0$, and taken a linear functional $\xi$: $R^n \to R$ with $\ker\xi = \psi(N \cap U)$, the function $\xi \circ (\psi \circ \varphi)$: $S \to R$ has a singularity of type $A_k$ in $z$. We recall that a smooth function germ $f$: $(R^m, 0) \to (R, 0)$ is of type $A_k$ if and only if it is equivalent, after a change of coordinates in the source, to a germ $g$ of the form:

$$g(x_1,\dots, x_m) = \pm x_1^2 \pm \dots \pm x_{m-1}^2 \pm x_m^{k+1}$$

**Definition 12** *We call* $\Sigma^*$ *a* regular net *for* $\{S, \varphi\}$ *if the projection* $\pi$: $S \to \Sigma$ *is* stable *and there is a stratification* $\Sigma^* = (\Sigma^*)_2 \cup (\Sigma^*)_1 \cup (\Sigma^*)_0$ *where each stratum* $(\Sigma^*)_i$ *is a smooth submanifold of dim.*i, *with a finite number of connected components, such that:*

$(\Sigma^*)_2 = \{$*all elements of* $\Sigma^*$ *which are transversal to* $\varphi(S)\}$;

$(\Sigma^*)_1 = \{$*all elements of* $\Sigma^*$ *having contact of type* $A_1$ *with* $\varphi(S)$ *at one point and transversal to* $\varphi(S)$ *at every other point*$\}$;

$(\Sigma^*)_0 = \{$*all elements of* $\Sigma^*$ *having contact of type* $A_2$ *with* $\varphi(S)$ *at one point, or contact of type* $A_1$ *at two different points, and transversal to* $\varphi(S)$ *at every other point*$\}$.

It follows from [Ma], with the same methods of proof, that given *any* embedding

$\varphi$: $S$ -> $RP^n$ a generic choice of the center of projection yields a stable map $\pi$: $S$ -> $\Sigma$. The hyperplanes of $\Sigma$ not transversal to $\varphi(S)$ represent the critical values of this map. They lie on a curve $\Delta$ displaying only cusps and double points [W2]. Let $Emb(S, RP^n)$ denote the set of all embeddings of $S$ in $RP^n$, endowed with the Whitney topology. A transversality argument, based on the results of [Lo] and exploiting the fact that for the simple singularities Looijenga's stratification coincides with the one defined in [T2, p.577], leads us to the following conclusion: *there is an open dense subset* $\mathcal{A} \subset \mathrm{Emb}(S, RP^n)$ *such that if* $\varphi \in \mathcal{A}$ *a generic choice of the center* C *gives rise to a regular net of hyperplanes* $\Sigma^*$ *for* $\{S, \varphi\}$.

From now on, we shall represent $\pi$ in terms of a two-dimensional hyperplane $H$ in $RP^n$, transversal to $C$. Every point of $H$ corresponds to an element of $\Sigma$: a curve $\gamma$ in $H$, with cusps and double points, is the locus of singular values of the projection. We shall refer to $\gamma$ as the *apparent contour* of $\varphi(S)$.

**Proposition 8** *If* $\Sigma^*$ *is a regular net of hyperplanes, its discriminant* $\Delta^*$ *is a curve with cusps and double points .*

***Proof*** Let's consider the dual curve $\Delta$, or rather its representation $\gamma \subset H$. Since $\Sigma^*$ is a regular net of hyperplanes, $\gamma$ cannot display any triple tangents (in a regular net $\Delta^* = (\Sigma^*)_1 \cup (\Sigma^*)_0$). We need to check that it has only ordinary inflections, and that no tangent of inflection can be a double tangent.

Let $p \in \gamma$ be a point of inflection, and let $q \in S$ be its inverse image in the component $\sigma_q$ of the fold curve which covers the branch of $\gamma$ displaying the inflection at $p$. Let's put ourselves in an affine chart containing $p$ and $\varphi(q)$ but not the center of the projection, and such that the elements of $\Sigma^*$ are seen in this chart as $(n - 1)$-hyperplanes orthogonal (according to the usual Euclidean scalar product) to the 2-plane determined by $H$. We may suppose that this 2-plane has coordinates $(x_1, x_2, 0,..., 0)$, with $p = (0, 0,..., 0)$, the $x_2$-axis being the tangent of inflection at $p$, and the $x_1$-axis the normal to the hyperplane of $\Sigma^*$ which contains the $x_2$-axis. We see that:

1) the "height function" $h$: $S$ -> R defined by the projection on the $x_1$-axis (let $C \subset \varphi(S)$ be the hyperplane section with the element of $\Sigma^*$ containing the $x_2$-axis, then $\varphi^{-1}(C)$ is the level curve $h^{-1}(0)$, near $q$) must have a degenerate critical point in $q$. This can be checked by looking at the topological type of the level curves $h^{-1}(\eta)$, when $\eta$ varies in a neighbourhood of 0.

2) Since, in the generic case, this singularity is of type $A_2$, the inflection point of $\gamma$ at $p$ must be an ordinary flex. In order to see this, we remark that the branch of $\gamma$ which displays the inflection at $p$ is of the form $x_1 = \psi(x_2)$, with $\psi(0) = \psi'(0) = \psi''(0) = 0$. Now, if $\xi(s)$ is a regular curve contained in $S$, with $\xi(0) = q$, the order of vanishing of $h(\xi(s))$ at $s = 0$ depends only on the tangent vector to $\xi(s)$ at $q$ (write $h$ in the form $x^2 + y^3$

after a suitable change of coordinates). One must have an order of vanishing < 3 in every direction, with the exception of one for which it is 3. This particular direction is that of the fold curve $\sigma_q$. By taking, near $q$, a chart of type $(x_2, t)$ we conclude that $h(x_2, 0) = \psi(x_2)$ must be such that $\psi'''(0) \neq 0$. □

Proposition 8 ensures that if the embedding $\varphi$ and the center $C$ are both chosen in a generic way, $\Delta^*$ and $\Delta$ (or $\gamma$) will be curves with cusps and double points, having a finite number of ordinary inflections and of double tangents (none of them being tangent to an inflection point), no triple tangents, etc. In particular, they are *convex-generic*.

Now suppose $\Sigma^*$ is a linear net of hyperplanes in $RP^n$, of center $C$, which is *regular* for $\{S, \varphi\}$, and let $\Delta^* = (\Sigma^*)_1 \cup (\Sigma^*)_0$ be its *discriminant*. Let $\mu$ and $\sigma$ be generators for $H^1(RP^2; Z_2)$ and $H^2(RP^n; Z_2)$, respectively. We indicate with $<\sigma, \varphi(S)>$ and $<\mu, \Delta^*>$ their Kronecker product with the homology classes induced by $\varphi(S)$ in $H_2(RP^n; Z_2)$ and by $\Delta^*$ in $H_1(RP^2; Z_2)$. Furthermore, let $w_2(S)$ be the *second Stiefel-Whitney number* of $S$, obtained by making the Kronecker product between the second Stiefel-Whitney class $w_2$ of the tangent bundle and the fundamental class of $S$.

**Theorem 2** *For a generic embedding* $\varphi: S \to RP^n$ *and a regular net* $\Sigma^*$, *we can compute* $<\mu, \Delta^*>$ *in the following way*:

$$<\mu, \Delta^*> = <\sigma, \varphi(S)> + w_2(S)$$

***Proof*** Let $\Sigma$ be the family of all $(n - 2)$-hyperplanes containing $C$ (that is, $\Sigma$ is the dual space of $\Sigma^*$). Let's consider the projection of center $C$; as usual, we represent $\Sigma$ and $\Delta$ by means of a two-plane $H$, in which $\Delta$ is embodied as the apparent contour of $S$, labelled by $\gamma$. Following our customary notations, we indicate with $\nu$ the normal bundle of $\gamma$ and with $m$ the primitive function that is canonically associated with the orientation of $\nu$ determined by the projection $f: S \to H$ (Definitions 8 and 9). Since $\gamma$ is *convex-generic*, $\chi_\gamma$ (Definition 7) is defined on all of $H$, and it represents the other "canonical" primitive function induced by the oriented apparent contour. We have seen these two primitives belong to different equivalence classes, which implies that their sum is constant (Corollary 1). In the generic setting we have ensured, this constant coincides with the Euler-Poincaré characteristic $\chi(S)$ of the surface (Proposition 7). Therefore the following relation holds:

$$\chi_\gamma + m = \chi(S)$$

Now we fix a point $p$ belonging to $\{H \setminus \gamma\}$. We draw all the tangents from $p$ to $\gamma$ and denote their number by $\alpha_p$. From the above relation, we immediately get:

$$\alpha_p = \chi(S) + m(p) \quad (\text{mod.}2)$$

Let $r$ be the line, contained in $\Sigma^*$, which is dual to the element of $\Sigma$ represented by $p$. We call $m_r$ the number of points at which this line intersects $\Delta^*$. Since $m_r = \alpha_p$, we obtain a dual formula:

$$m_r = \chi(S) + m(p) \quad (\text{mod.}2)$$

Let $T \in \Sigma$ be the $(n - 2)$-hyperplane containing $p$ and $C$. $T$ and $\varphi(S)$ lie in general position and $m(p)$ tells us how many times the embedded surface $\varphi(S) \subset RP^n$ intersects $T$. We observe that, by applying Poincaré duality:

1) $m_r$ is *even* $\Leftrightarrow$ $<\mu, \Delta^*> = 0$

2) $m(p)$ is *even* $\Leftrightarrow$ $<\sigma, \varphi(S)> = 0$

We recall that the second Stiefel-Whitney number $w_2(S)$ of $S$ is equal to the Euler-Poincaré characteristic of the surface, reduced mod.2. This final remark concludes the proof of Theorem 2. □

Theorem 2 can be used in order to extend to mappings to the *projective plane* some classical results on the number of cusps of a generic map. The theorems available in the literature concern maps taking their values in a Euclidean plane [W2, T1, H], as well as in any *orientable* surface [L1, E] (a more general relation is quoted, without proof, in [F-I]). So, let $S$ be a compact connected surface without boundary: we are going to see how the second Stiefel-Whitney class $w_2$ of the tangent bundle of $S$ is reflected by the global structure of a generic (stable) smooth mapping $f: S \to RP^2$. More precisely, we will prove that $w_2$ is dual to the 0-cycle defined by the cusps and by a generic fiber $f^{-1}(p)$ of $f$.

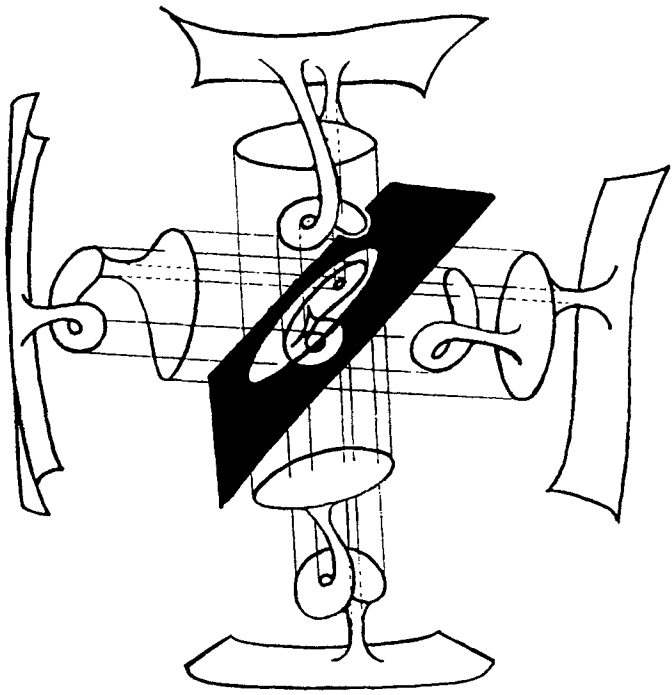
**Corollary 2** *Let* $d^*(f)$ *denote the mod.2 degree of* f, $c^*$ *the number of cusps (mod.2), and* $\chi^*(S)$ *the mod.2 Euler-Poincaré characteristic of the surface. Then :*

$$c^* = d^*(f) + \chi^*(S)$$

**Proof** We embed $RP^2$ as a projective subspace of $RP^n$, $n \geq 5$. Let $j: RP^2 \to RP^n$ be this embedding and set $H = \mathrm{Im} j$. We fix a center of projection $P \in RP^n$ (an $(n - 3)$-hyperplane $P$ not intersecting $H$) and consider the map $\pi: \{RP^n \setminus P\} \to H$ introduced before. Let $N$ be a neighbourhood of $f$ in the space of smooth mappings from $S$ to $RP^2$, chosen in such a way that $\forall\, g \in N$, $g$ is stable and smoothly equivalent (and homotopic) to $f$. The map $j \circ f$ can be approximated by a generic mapping $f': S \to RP^n$ belonging to the class $\mathcal{A}$ of embeddings introduced at the beginning of this section, and having the property that $\pi \circ f' \in N$. This ensures that the number $c$ of the cusps in the apparent contours of $f$ and $\pi \circ f'$ is the same. We can take a new center of projection $C$ close to $P$, so that the corresponding map $\pi' \circ f'$ approximates $\pi \circ f'$ arbitrarily well and its apparent contour $\gamma'$ is convex-generic, has only ordinary inflections, etc. (Proposition 8). Since $\pi \circ f'$ is a stable mapping, the number of cusps in the apparent contour will keep the same.

Let $\Delta^*$ be the discriminant of the net $\Sigma^*$ defined by the new center $C$ and intersecting $f'(S)$. We have (Theorem 2): $<\mu, \Delta^*> = <\sigma, f'(S)> + w_2(S)$.

We observe that: $<\mu, \Delta^*> = 0 \Leftrightarrow$ the number of irreducible components of $\Delta^*$ not homologous to zero is even $\Leftrightarrow$ the number of flexes of $\Delta^*$ is even (Proposition 2).



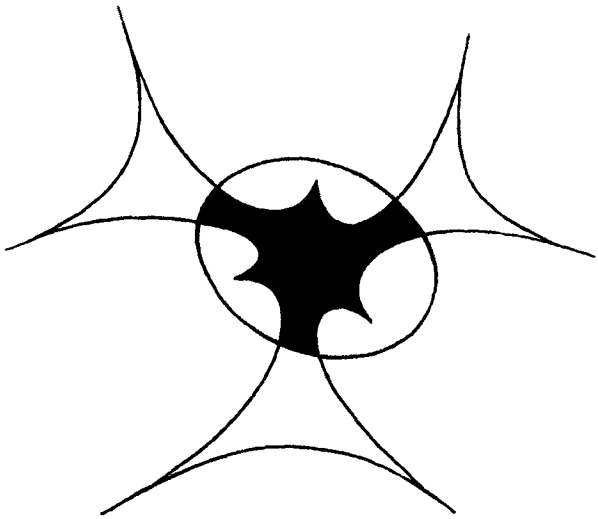


Fig.12

The number of flexes of $\Delta^*$ is equal to the number $c$ of cusps of $f$; in addition, for $f'$ close to $j\circ f$, the two cycles $j\circ f(S)$ and $f'(S)$ lie in the same homology class. Now recall that we denoted by $c^*$ the number (mod.2) of cusp points of $f$. Since $w_2(S) = \chi^*(S)$, $c^*$ verifies the following relation: $c^* = \langle\sigma, j\circ f(S)\rangle + \chi^*(S)$.

The homology class defined by $j\circ f(S)$ in $H_2(RP^n; Z_2)$ is zero if and only if the mod.2 degree of $f$ is zero, and this remark concludes the proof. □

***Remark* 2** In order to realize the meaning of the relation we have just proved, we can think of some examples of surfaces embedded in $RP^3$ and projected to an hyperplane $H$. Let $g$ be the genus of $S$ and denote by $[S]$ the homology class defined by the surface in $H_2(RP^3; Z_2)$. Let's put ourselves in an affine chart and look at the contour. For *orientable* surfaces, the preceding result says that $c$ must be *even*. In fact, we have already remarked that a two-sheeted hyperboloid, standardly embedded and projected in the direction of its axis, engenders an empty contour. A one-sheeted hyperboloid produces a circle. In both cases $c$ is "even", being equal to zero.

Turning to the *non-orientable* case we have: for $g$ *odd*, $[S] = 0 \Leftrightarrow c$ is odd; for $g$ *even* $[S] = 0 \Leftrightarrow c$ is even. Fig.1 gives some examples with $[S] = 0$. To get an example with $[S] \neq 0$, just remember that $RP^2$, embedded as an hyperplane of $RP^3$, has an empty contour. Hence $g$ is odd, $c$ even. Other examples for $RP^2$ are given in Fig.12 (top): we get the same contour that was seen in Fig.8, but in the present case $g = 1$ (odd), $c = 2$ (even) and $[S] \neq 0$. To the bottom of Fig.12 we see the contour of a surface of genus 6, embedded in $RP^4$. It is obtained by joining a two-dimensional hyperplane, by means of three handles, to another projective plane. The last is embedded in an affine chart, just like a generic 4-dimensional shadow of the Veronese surface [Ba, Fig.3], whose projection to the plane gives rise to the same familiar contour as the "Roman surface" [A, F3] or Boy surface's "flat view" [F-T1, Fig.3.2]. Here $g$ is even, $c$ odd and $S$ is not homologous to zero.

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Roberto Pignoni
Dip. di Matematica "Federigo Enriques"
Università di Milano
Via C.Saldini n.50, 20133 Milano, Italia

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